

Lingam Hemang

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Professional Summary

Senior Python Backend Engineer with **4 years** at Infosys building **production backend systems, REST APIs, and GenAI applications** for **Apple Inc.** Delivered **40% faster pipelines, 99.5% uptime, 35% fewer errors** on a large-scale distributed platform built on **Python, Spark, Kafka, and AWS**. Strong expertise in **Django, Flask, OOP, SOLID principles, and API design**. Actively building GenAI applications using **LangChain, RAG pipelines, and AWS Bedrock**.

Technical Skills

Languages: Python, SQL, PySpark

Backend & APIs: Django, Flask, REST API Design & Integration, Server-Sent Events (SSE), OOP, SOLID Principles, Design Patterns, pytest, Microservices

Data Engineering: Apache Spark, Apache Kafka, Apache Airflow, Snowflake, AWS S3, AWS Lambda, ETL Pipelines, Data Modelling, Schema Design & Validation, Partitioning & Query Optimisation

GenAI / ML: AWS Bedrock, LangChain, RAG Pipelines, Vector Embeddings, ChromaDB, XGBoost, LightGBM, scikit-learn, Prompt Engineering

DevOps & Tools: Docker, Git, CI/CD, Linux, Bash, Splunk, Agile/Scrum, Jira, Confluence

Experience

Infosys

Hyderabad, India

Specialist Programmer – Senior Python Backend Engineer

July 2022 – Present

Client: Apple Inc. | Python, Django, Flask, Spark, Kafka, Airflow, Snowflake, AWS S3, AWS Lambda, REST APIs

- Built and maintained **Python backend services** consuming internal and external **REST APIs** (in-house endpoints, Pharos) to ingest and process data across a large-scale **distributed platform**; **SQL query optimisation and distributed processing** cut end-to-end runtime by **40%**.
- Designed **high-availability, fault-tolerant data pipelines** across **AWS S3** and **Kafka event streams** via **Airflow DAGs** with retry logic and SLA monitoring – achieving **99.5% uptime**; **schema validation and self-healing scripts** reduced error rate by **35%**.
- Developed **reusable Python framework extensions** for new data source integrations, cutting onboarding effort by **30%**; built **semantic layer processes** with materialised views for downstream analytics; created a **Tableau security audit dashboard** to monitor data access patterns.
- Provided **backend data support** for a production application – maintaining **database view consistency** and loading data into a **semantic search cluster**; managed releases via **CI/CD** pipelines; collaborated across Agile feature teams via Jira and Confluence.

Projects

Agentic AI Git Diff Analyzer | Python, Django, AWS Bedrock, AWS Lambda, Ollama, ReportLab | [Repo](#)

- Dual-mode **SQL schema impact analyzer** with **Django REST backend** – local (Ollama) and AWS mode (Bedrock Claude 3.5 via GitHub Webhook → Lambda → EC2); LLM-generated impact reports as styled PDFs; real-time **SSE streaming** with commit-level deduplication.

Document Intelligence Chatbot – RAG Pipeline | Python, Django, LangChain, ChromaDB, Ollama, EasyOCR | [Repo](#)

- 10-endpoint Django REST API** powering an end-to-end **RAG pipeline** (LangChain + ChromaDB + Ollama); multi-format ingestion via PyMuPDF + EasyOCR; persistent multi-turn conversation memory; live model switching and VectorDB management endpoints.

Retail ML Sales Forecasting API | Python, Flask, scikit-learn, XGBoost, Plotly, Dash | [Repo](#) **Infosys Hackathon – 2nd Place**

- Multi-model **ML forecasting system** (Random Forest, XGBoost, LightGBM) via **Flask REST API**; cyclical sin/cos encoding for seasonal patterns; cross-join purchase probability scoring; interactive **Dash + Plotly** dashboard with live reactive callbacks.

Education

SRM Institute of Science and Technology

May 2018 – June 2022

Bachelor of Technology – Electronics and Communication Engineering

CGPA: 9.52/10.0